

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – *Research Paper Review / Literature Survey*

Course Code:	CSA1013	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	SPANDANA S	Reg. No.:	192525160
Faculty In-Charge:	Latha	Date:	24-08-2026
Duration:	60 minutes		

Code of Conduct

I SPANDANA S / 192525160 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, conduct a literature survey and review relevant research papers to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select relevant research papers related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the objectives, methodologies, technologies, datasets, and major findings of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics.
4. Identify the limitations and challenges of the existing research approaches.
5. Analyze the literature to identify the research gap or unresolved issues related to the assigned problem.
6. Discuss the possible improvements or future research directions based on the identified research gap.
7. Prepare a comparative literature survey table containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper citations and references using a consistent referencing style.

Deliverable: Submit a Research Paper Review / Literature Survey Report containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers (25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis (20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap (25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance (15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References (15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Marks Awarded

Criteria	Mark
Selection & Review of Research Papers (25%)	/5
Comparative Analysis (20%)	/5
Critical Analysis & Research Gap (25%)	/5
Future Directions & Relevance (15%)	/5
Documentation, Citations & References (15%)	/5
TOTAL	/25

RESEARCH PAPER REVIEW / LITERATURE SURVEY REPORT

Topic 38: Intelligent Smart Port Operations and Cargo Management Platform

Student Name	SPANDANA S
Registration Number	192525160
Course Code / Title	CSA1013 – Software Engineering
Department	Computer Science and Engineering
Institution	SIMATS Engineering (Saveetha School of Engineering)
Faculty In-Charge	Latha
CO Assessed	CO5 – Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly

1. Introduction and Industry Context

Modern seaports are among the busiest nodes in global trade, handling large and continuously growing volumes of cargo and vessel traffic every day. Coordinating this scale of activity requires efficient logistics planning and real-time operational monitoring across berths, yards, gates, and warehouses. The emergence of the Internet of Things (IoT), Radio Frequency Identification (RFID), Global Positioning System (GPS), Artificial Intelligence (AI), and cloud computing has enabled ports to move away from manual, paper-based coordination toward intelligent, data-driven operations.

These technologies allow real-time cargo tracking, automated vessel and berth scheduling, and predictive analytics that anticipate delays before they occur. Together they form the foundation of the “smart port” concept, in which physical infrastructure is tightly integrated with digital systems to improve productivity, reduce operational delays, and support transparent, secure port operations.

2. Problem Statement

Port authorities frequently experience operational inefficiencies caused by cargo congestion, manual documentation, delayed vessel scheduling, and limited visibility of cargo movement. These challenges increase vessel turnaround time and overall logistics costs. An intelligent smart port management platform is required to monitor cargo movement using IoT devices, optimize berth allocation, predict cargo handling delays using AI, automate documentation processes, and provide real-time operational dashboards. The system should enhance logistics efficiency while supporting secure and transparent port operations.

3. Objectives

1. Monitor cargo movement in real time.
2. Optimize berth and vessel scheduling.
3. Predict cargo handling delays using AI.
4. Automate logistics documentation.
5. Improve warehouse coordination.
6. Generate operational performance reports.
7. Enhance cargo security.
8. Increase port operational efficiency.

4. Literature Survey – Individual Paper Reviews

Six research papers relevant to the assigned problem statement were identified from IEEE Xplore, Springer, PLOS ONE, MDPI Sensors, and a peer-reviewed marine science journal. They collectively cover the core building blocks of a smart port and cargo management platform: IoT-based cargo and vessel tracking, AI-based vessel traffic counting, machine learning-based berth allocation and vessel arrival prediction, and blockchain-based logistics documentation.

4.1 S. Priya, A. Sairam, H. Azath, S. Manivannan, N. Mohankumar, and P. V. Vinayagam, “Smart Ports Solutions for Cargo Container Tracking and Vessel Traffic Counting Systems using IoT and Cloud Computing,” in *Proc. 2024 10th Int. Conf. on Communication and Signal Processing (ICCSP)*, IEEE, 2024, pp. 1106–1111.

Objective	To design an IoT-based framework for real-time cargo container monitoring and vessel traffic counting to reduce port congestion.
Methodology / Technology	IoT sensors and devices attached to cargo containers report location, condition, and security status; cloud computing aggregates and analyzes this data alongside sensor-based vessel traffic counts.
Dataset / Tools	Live IoT sensor and device data streams from port container yards and vessel traffic points; cloud-based data analytics platform.
Key Findings	Real-time IoT tracking improved logistical planning and cargo security, and sensor-based vessel counting reduced congestion and vessel waiting time at key port locations.

4.2 M. Petković, I. Vujović, N. Kaštelan, and J. Šoda, “Every Vessel Counts: Neural Network Based Maritime Traffic Counting System,” *Sensors*, vol. 23, no. 15, p. 6777, 2023.

Objective	To build a camera-based, AI-driven vessel counting and classification system as a low-cost complement to AIS-based vessel traffic monitoring.
Methodology / Technology	A YOLOv4 convolutional neural network classifies vessels into twelve categories from land-based camera footage, combined with a Kalman tracker and Hungarian Assignment multi-target tracking algorithm.
Dataset / Tools	The purpose-built SPSCD video dataset collected at the Port of Split, Croatia.
Key Findings	The system achieved an average counting accuracy of 97.76% at 31.78 frames per second and captured 386% more maritime traffic data than conventional AIS systems.

4.3 M. H. Alshareef, “Machine Learning Applications for Berth Allocation Optimization in Major Red Sea Ports: A Multi-Port Analysis,” *Journal of King Abdulaziz University: Marine Science*, vol. 35, no. 1, Art. 9, 2025.

Objective	To apply deep learning to the berth allocation problem across multiple Red Sea ports, incorporating environmental and operational constraints.
Methodology / Technology	A deep learning model with four dense layers and ReLU activation functions was trained on historical berth allocation records and validated against traditional allocation methods.
Dataset / Tools	147,892 historical berth allocation instances (2019–2023) across several Red Sea ports, including vessel characteristics and weather patterns.
Key Findings	The model reduced vessel waiting time by 33.2%, improved berth utilization by 27.7%, and cut berthing-related fuel consumption by 21.4% compared with conventional allocation methods.

4.4 A. Albloushi et al., “Predicting Vessel Arrival Time Using Machine Learning for Enhanced Port Efficiency and Optimal Berth Allocation,” in *Proc. Int. Conf. (Springer)*, 2025.

Objective	To predict vessel estimated arrival times accurately enough to support optimal berth allocation decisions and reduce port congestion.
Methodology / Technology	Three regression models — Gradient Boosting, K-Nearest Neighbors, and Random Forest — were trained and compared on real-world vessel movement data.
Dataset / Tools	Historical AIS data from the Norwegian Base Station and satellite observations collected

	between 1 August and 24 September 2024.
Key Findings	The Random Forest model performed best, achieving an R^2 of 0.704 and a mean absolute percentage error of 0.0285%, demonstrating strong potential for berth-scheduling support.

4.5 L. Kolley, N. Rückert, M. Kastner, C. Jahn, and K. Fischer, “Robust Berth Scheduling Using Machine Learning for Vessel Arrival Time Prediction,” Flexible Services and Manufacturing Journal, vol. 35, no. 1, pp. 1–32, 2023.

Objective	To derive conflict-free, robust berth schedules that explicitly account for uncertainty in vessel arrival times caused by wind and sea current.
Methodology / Technology	Four machine learning methods, ranging from linear regression to an artificial neural network, were used for vessel arrival time prediction and integrated into a data-driven berth scheduling optimization.
Dataset / Tools	Historical terminal operating data on vessel arrivals, quay assignments, and prevailing weather conditions.
Key Findings	Incorporating machine learning-based arrival predictions into the scheduling model produced measurably more robust, conflict-free berth schedules than deterministic scheduling alone.

4.6 M. Liaqat, A. A. Almazroi, J. Shuja, and E. Mustafa, “Securing Oil Port Logistics: A Blockchain Framework for Efficient and Trustworthy Trade Documents,” PLOS ONE, vol. 19, no. 10, e0309526, 2024.

Objective	To eliminate manipulation and delay in port trade documentation (e.g., bills of lading, customs declarations) by automating and securing the process with blockchain.
Methodology / Technology	A private, permissioned blockchain built on Hyperledger Fabric, with role-based smart contracts governing document creation, verification, updating, and delivery among port stakeholders.
Dataset / Tools	Simulation of 1,000 blockchain transactions on an Ubuntu test network with CouchDB ledger storage, evaluated for latency, throughput, and resource utilization.
Key Findings	The framework provided tamper-proof, role-restricted document handling with measurable throughput (up to 1.67 transactions/second) and demonstrated transparency, immutability, and trust among stakeholders.

5. Comparative Literature Survey Table

The table below compares the six selected studies on research problem, methodology, dataset/tools, key findings, limitations, and the specific research gap each leaves open with respect to a unified smart port platform.

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Priya et al., 2024	Real-time cargo container monitoring and vessel traffic congestion	IoT sensors + cloud computing analytics	IoT device data streams (container yard, vessel traffic points)	Improved logistical planning, cargo security and traffic flow	Data privacy and security risks; heavy dependence on IoT infrastructure investment	No predictive or AI-based delay forecasting layer
Petković et al., 2023	Incomplete vessel traffic visibility from AIS/VTIS alone	YOLOv4 CNN + Kalman/Hungarian multi-target tracking	SPSCD camera dataset, Port of Split	97.76% counting accuracy at 31.78 fps; 386% more traffic data than AIS	Camera coverage and lighting/weather sensitivity; limited to visual range	Not integrated with cargo tracking or berth scheduling systems
Alshareef, 2025	Inefficient berth allocation across multiple ports	Deep learning (4 dense layers, ReLU)	147,892 berth records (2019–2023), Red Sea ports	33.2% less waiting time; 27.7% better berth utilization; 21.4% less fuel use	Regional dataset (Red Sea only); limited generalizability to other port types	Lacks integration with real-time cargo and documentation modules
Albloushi et al., 2025	Inaccurate vessel arrival time estimates disrupting berth planning	Gradient Boosting, KNN, Random Forest regression	AIS data, Norwegian Base Station (Aug–Sep 2024)	Random Forest best model: $R^2 = 0.704$, MAPE = 0.0285%	Single-region, short time-window dataset; no live operational deployment	Prediction not yet coupled with automated berth-assignment decisions

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Kolley et al., 2023	Berth schedule conflicts from uncertain vessel arrival times	Comparative ML models (linear regression to ANN) + robust scheduling	Terminal arrival, quay and weather records	More robust, conflict-free berth schedules than deterministic methods	Model accuracy depends on quality/availability of historical terminal data	Limited scalability testing across multiple terminals simultaneously
Liaqat et al., 2024	Slow, error-prone, manipulable port trade documentation	Permissioned blockchain (Hyperledger Fabric) + smart contracts	1,000 simulated transactions, Ubuntu test network	Tamper-proof records; throughput up to 1.67 transactions/sec	High transaction latency (~600 s per batch); not tested at full-scale port volume	No integration with IoT cargo sensors or AI-based scheduling components

6. Limitations and Challenges of Existing Approaches

- IoT-based tracking systems (Priya et al., 2024) raise data privacy and security concerns and require significant sensor and network infrastructure investment.
- Camera-based AI vessel counting (Petković et al., 2023) is sensitive to weather, lighting, and camera coverage, and works only within visual range.
- Machine learning-based berth allocation models (Alshareef, 2025; Kolley et al., 2023) are trained and validated on data from specific regions or terminals, limiting generalizability to ports with different traffic patterns.
- Vessel arrival prediction models (Albloushi et al., 2025) are typically evaluated over short time windows and narrow geographic coverage, and are rarely tested in live operational settings.
- Blockchain-based documentation frameworks (Liaqat et al., 2024) show notable transaction latency at scale and have not been tested under full port-level transaction volumes.
- Across all six studies, each technology (IoT, computer vision, ML prediction, blockchain) is developed and evaluated in isolation, with no single study integrating cargo tracking, berth optimization, delay prediction, and documentation automation into one unified platform.

7. Research Gap

The reviewed literature demonstrates strong individual progress in IoT-based cargo and vessel tracking, computer-vision-based traffic counting, machine learning-based berth allocation and arrival prediction, and blockchain-based trade documentation. However, no reviewed study combines these components into a single, unified smart port and cargo management platform that simultaneously monitors cargo in real time, optimizes berth and vessel scheduling, predicts cargo handling delays, automates documentation, and presents this information through a consolidated operational dashboard.

Further, most of the reviewed machine learning and deep learning models are validated on data from a single port or a narrow geographic region and over comparatively short observation periods, which raises open questions about their robustness and generalizability when deployed across ports with different scales, cargo mixes, and environmental conditions. Similarly, blockchain-based documentation frameworks have not yet been evaluated at full port-level transaction volumes, leaving their scalability under real operational load largely unaddressed. This represents the core research gap that the assigned problem statement – an intelligent smart port operations and cargo management platform – seeks to address.

8. Future Research Directions and Possible Improvements

- Develop a unified middleware layer that integrates IoT cargo/vessel sensor data, computer-vision-based traffic counting, ML-based delay and arrival prediction, and blockchain-based documentation within a single operational platform.
- Validate berth allocation and vessel arrival prediction models across multiple ports and longer time periods to improve generalizability beyond a single region.

- Apply reinforcement learning to jointly optimize berth allocation, quay crane assignment, and yard operations rather than treating each as a separate optimization problem.
- Benchmark blockchain documentation frameworks at full port-scale transaction volumes and explore hybrid or layer-2 approaches to reduce transaction latency.
- Incorporate real-time environmental and weather data more systematically into predictive models, given the demonstrated impact of weather on berthing and arrival time accuracy.
- Design real-time operational dashboards that fuse cargo visibility, berth/vessel status, predicted delays, and documentation status into a single decision-support view for port authorities.

9. Expected Outcomes

Based on the assigned problem statement and the reviewed literature, an integrated smart port and cargo management platform is expected to deliver the following outcomes:

9. Reduced cargo handling time.
10. Improved logistics efficiency.
11. Better vessel scheduling.
12. Enhanced cargo visibility.
13. Lower operational costs.
14. Improved port productivity.
15. Increased supply chain reliability.
16. Better customer service.

10. Conclusion

This literature survey reviewed six recent research papers addressing the core technological pillars of smart port operations: IoT-based cargo and vessel tracking, AI-based vessel traffic counting, machine learning-based berth allocation and arrival time prediction, and blockchain-based trade documentation. Each study demonstrates measurable operational gains — from congestion reduction and improved counting accuracy to reduced vessel waiting time and tamper-proof documentation — but each also addresses only one part of the smart port problem in isolation.

The comparative analysis and research gap identified in this survey confirm the need for an intelligent, integrated smart port and cargo management platform that combines real-time cargo monitoring, AI-driven berth and delay prediction, automated documentation, and consolidated operational dashboards, as called for in the assigned problem statement. Addressing this gap would directly support the expected outcomes of reduced cargo handling time, improved logistics efficiency, and increased supply chain reliability.

11. References

- [1] S. Priya, A. Sairam, H. Azath, S. Manivannan, N. Mohankumar, and P. V. Vinayagam, “Smart Ports Solutions for Cargo Container Tracking and Vessel Traffic Counting Systems using IoT and Cloud Computing,” in Proc. 2024 10th Int. Conf. on Communication and Signal Processing (ICCSP), IEEE, 2024, pp. 1106–1111.
- [2] M. Petković, I. Vujović, N. Kaštelan, and J. Šoda, “Every Vessel Counts: Neural Network Based Maritime Traffic Counting System,” *Sensors*, vol. 23, no. 15, p. 6777, 2023.
- [3] M. H. Alshareef, “Machine Learning Applications for Berth Allocation Optimization in Major Red Sea Ports: A Multi-Port Analysis,” *Journal of King Abdulaziz University: Marine Science*, vol. 35, no. 1, Art. 9, 2025.
- [4] A. Albloushi et al., “Predicting Vessel Arrival Time Using Machine Learning for Enhanced Port Efficiency and Optimal Berth Allocation,” in Proc. Int. Conf. (Springer), 2025.

- [5] L. Kolley, N. Rückert, M. Kastner, C. Jahn, and K. Fischer, “Robust Berth Scheduling Using Machine Learning for Vessel Arrival Time Prediction,” *Flexible Services and Manufacturing Journal*, vol. 35, no. 1, pp. 1–32, 2023.
- [6] M. Liaqat, A. A. Almazroi, J. Shuja, and E. Mustafa, “Securing Oil Port Logistics: A Blockchain Framework for Efficient and Trustworthy Trade Documents,” *PLOS ONE*, vol. 19, no. 10, e0309526, 2024.